

Med Priority Process started working at : 1522160702
My Priority is : 28

Medium Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./low
Low process arrived at : 1522160686
Press any button to enter into critical section
```

```
start Time: 1522160699
Low Priority Process started working at : 1522160699
My Priority is : 20
My Priority is : 20
endTime: 1522160719
Process completed
```

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test#
```

Low Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./high
High process arrived at : 1522160693
Press any button to enter into critical section
```

```
High process waiting for lock start Time: 1522160719
High Priority Process started working at : 1522160719
```

```
My Priority is : 38
My Priority is : 38
endTime: 1522160724
Process completed
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test#
```

High Priority Process

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test# ./med
```

Med Priority Process arrived at: 1522160696
Press any button to start work...

start Time: 1522160702

```
Med Priority Process started working at : 1522160702
My Priority is : 28
My Priority is : 28
endTime: 1522160722
Process completed
```

```
root@dhanashree-Inspiron-5559:/media/dhanashree/New Volume/
Study/sem2/ProjectElective/Final Test#
```

Medium Priority Process

Using PI-Futex for locking the critical section removes the problem of priority inversion—by inheriting the priority of the low priority process in the critical section when a high priority process waits on futex. Along with processes, futexes can be used for threads also. Futex is comparatively faster than semaphore, as the latter involves the kernel in both contended and non-contended cases, whereas futex involves the kernel only in contended cases. Nowadays, almost all systems have multi-core processors; so by forcing processes to run on different processors, the priority inversion problem can be avoided. **END** 🐧

References

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